

Bijan Pourriahi

Senior Software Engineer | Developer Platforms & APIs

Rust · Python · Integrations · AI Tooling · Production Systems

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PROFILE

Senior Software Engineer with 10+ years of experience building production software across developer-facing platforms, API systems, integrations, runtime tooling, and operational infrastructure. Strong hands-on background in Rust, Python, API design, WebSockets, external API integrations, CI/CD, Linux, monitoring, debugging, and production maintenance.

I build systems that make complex workflows easier to develop, debug, operate, and trust: clear interfaces, fast feedback loops, reproducible execution, useful observability, integration adapters, orchestration layers, and AI-assisted developer workflows. Recent work includes Rust/Python platform infrastructure where I was the primary user, so developer experience, debugging speed, and operational visibility directly shaped the architecture.

SELECTED ENGINEERING WORK

Developer platform engineering: Built Rust/Python platform components focused on clear interfaces, reproducible workflows, fast debugging, and operational visibility for complex execution systems.

APIs and integrations: Built API and integration layers across external systems with different reliability characteristics, including WebSocket services, authentication flows, broker/execution APIs, market-data providers, CLIs, dashboards, and internal tooling.

Workflow orchestration: Built scheduled workflows, replay systems, evaluation pipelines, and operational tooling with emphasis on reproducibility, observability, and maintainability.

AI tool interfaces: Built MCP-based interfaces and CLI tooling for AI-assisted interaction with internal services, choosing appropriate interfaces based on workflow requirements.

Production systems: Owned deployment, monitoring, incident response, release workflows, and operational tooling for reliable production systems.

PROFESSIONAL EXPERIENCE

Private Financial Systems Client — Contract Senior Software Engineer / Technical Lead, Platform & Runtime Systems

2021-Present | Remote

- Owned architecture, implementation, deployment, and operation of a proprietary Rust/Python platform that made complex research and execution workflows reproducible, debuggable, observable, and easier to operate.
- Designed shared Rust/Python research and runtime components so historical backtests and live strategy evaluation could use the same core semantics, including replay, simulation engines, account-state models, risk controls, broker-integration logic, and evidence-capture tooling.
- Built API and integration layers for brokers, prop firms, market-data providers, execution APIs, WebSocket APIs, authentication flows, and external systems with different latency, reliability, and failure-mode characteristics.
- Built MCP interfaces and CLI tools for AI-assisted access to internal services, choosing the appropriate tool abstraction based on whether the workflow needed remote exposure, local simplicity, standardization, or easier debugging.
- Built WebSocket command/control flows, operator dashboards, account-state tracking, risk-control workflows, and monitoring surfaces for production use across 10+ live client trading accounts.
- Built workflow orchestration systems for scheduled experiments, replay jobs, model/strategy evaluations, result capture, and operational review, after using Airflow, Dagster, and Argo Workflows on earlier workflow systems.
- Built SQL-backed analytics and experiment-review tools across PostgreSQL, ClickHouse, and SQLite so operators could compare variants, inspect metrics, review evidence, and reproduce prior decisions.
- Evaluated thousands of strategy/model configurations using XGBoost, Random Forests, and simpler statistical models to test whether market conditions, engineered features, and strategy parameters improved decision quality.
- Built deployment, monitoring, alerting, health-check, and operational review workflows using Git, Docker, AWS/Linux environments, Grafana, Prometheus, and production runbooks.

Rare Isle Fund — Principal Software Engineer

Apr 2019-Jun 2021 | Remote

- Principal engineer for a small hedge fund, responsible for core automation, analytics, dashboards, data pipelines, and production operations.
- Built automated trading, research, analytics, and operations infrastructure from scratch, replacing manual research and strategy review with repeatable software workflows.
- Designed Python/Pandas/SQL pipelines for ingesting market data, generating research datasets, running high-volume parameter sweeps and historical simulations, storing results, ranking strategy variants, and reviewing statistical performance.
- Built operational controls, analytics dashboards, monitoring tools, and automated review systems to improve decision speed, production visibility, and system reliability.
- Owned implementation and operations across Python, Pandas, Go, SQL, AWS, Grafana, Linux, DevOps, data pipelines, and production infrastructure.

FocusVision — Senior Software Engineer

Nov 2015-Jun 2018 | Portland, OR

- Senior engineer in a market research SaaS company with multiple acquired production systems and integration needs.
- Led engineering work to integrate five acquired companies/systems into FocusVision, consolidating Rails/PostgreSQL SaaS systems into a more unified production platform.
- Led and mentored engineers while delivering SaaS features, integration work, APIs, database-backed workflows, bug fixes, and production improvements.
- Managed deployments and coordinated with sysadmin/operations teams across production releases, infrastructure changes, data migrations, integration work, and customer-impacting fixes.
- Owned architecture and delivery across requirements analysis, backend/application changes, testing, debugging, documentation, production support, and cross-team coordination.

ADDITIONAL EXPERIENCE

24tru — Software Engineer

Apr 2013-Jun 2015 | Lakewood, CO

- Worked in a small market research SaaS team building software for research collection, data processing, search, and video-based workflows.
- Co-owned the company software platform with one other developer, covering production maintenance, data migrations, new features, customer-impacting fixes, and day-to-day technical operations.
- Managed deployment of the company software stack from the ground up, including release processes, production maintenance, and operational reliability.
- Improved search/Solr indexing performance by roughly 100x by replacing a slow indexing workflow with a custom Java indexing application.
- Built APIs and product features that helped customers collect, upload, process, transcribe, search, and manage video research data.

gloo — Software Developer

Feb 2012-Oct 2012 | Boulder, CO

- Worked in a digital agency/client-services environment.
- Implemented Ruby on Rails features, fixed bugs, wrote tests, and supported PostgreSQL-backed client product development.

TECHNICAL STACK

Languages: Rust, Python, SQL, Ruby, Go, TypeScript, Java

Systems / APIs: API design, WebSockets, REST integrations, command/control flows, CLIs, internal tools, browser tools, GraphQL ecosystem, federation patterns

AI tooling: MCP servers/tools, Codex, Claude-style coding agents, agent workflow design, tool orchestration, AI-assisted debugging/review/documentation

Platform: Docker, AWS, Linux, Git, CI/CD, Grafana, Prometheus, Airflow, Dagster, Argo Workflows

Data: PostgreSQL, ClickHouse, SQLite, Pandas, event logs, replayable workflows, query optimization, experiment tracking